



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2024– 2025 (Odd Semester)

Degree, Semester & Branch : VII Semester B.E CSE
Course Code & Title : CCS346 & Exploratory Data Analysis
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit II/Univariate Analysis/Numerical summaries
- **Course Outcome:** CO3
- **Topic Learning Outcome:** 3b
- **Activity Chosen:** Problem Based Learning
- **Justification:**
 - ❖ This activity facilitates collaborative learning by engaging students in problem-solving tasks that relate to calculating and interpreting numerical summaries such as mean, median, mode, variance, and standard deviation in EDA.
 - ❖ Instead of direct instruction, students are given real-world data sets and asked to calculate these numerical summaries. They will need to identify the appropriate summary statistics based on the problem context, fostering a deeper understanding of the significance and application of these measures.
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**

After completing the second unit, an activity was designed to assess students understanding of numerical summaries and promote reflection on how these summaries are used to interpret data in EDA.

 - The class is divided into small groups of 4-5 students.
 - Each group is given a different real-world dataset.
 - Groups are tasked with calculating and interpreting key numerical summaries for their dataset (mean, median, mode, range, variance, standard deviation).
 - Using the provided dataset, calculate the mean, median, mode, and standard deviation.
 - Each group calculates the required statistics.
 - Students reflect on the implications of these summaries in relation to the dataset (e.g., if data is skewed or if outliers are influencing the summary statistics).
 - After calculating the statistics, groups categorize their insights based on key concepts like central tendency, dispersion, and data variability.
 - Groups present their findings and insights to the class, fostering discussion on how numerical summaries impact EDA interpretations.
 - Ms. K. Anitharani's team presenting their solution to the problem in class.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
CO1	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12
	3	3	3	3	3	3
Justification for correlation	Students apply comprehensive engineering knowledge to solve real-world problems, achieving a high level of engagement	Problem-based learning involves critical analysis and problem-solving, aligning perfectly with this outcomes	Students design and develop solutions to complex problems, indicating a high level of practical application	Student help in breaking down complex concepts into manageable parts, making it easier to communicate and collaborate within a team.	Written communication/ Asking questions are improved	Student will implement the questioning skill in future in written format

• **Images / Screenshot of the practice:**

Handwritten student work showing calculations for mean, variance, and standard deviation. The work is written on lined paper and includes the following steps:

$$\text{Mean} = \frac{4+6+5+3+9}{5}$$

$$= \frac{27}{5}$$

$$= 5.4$$

$$\text{Variance} = \frac{(4-5.4)^2 + (6-5.4)^2 + (5-5.4)^2 + (3-5.4)^2 + (9-5.4)^2}{5}$$

$$= \frac{1.96 + 0.36 + 0.16 + 5.76 + 12.96}{5}$$

$$= \frac{21.2}{5} = 4.24$$

$$\text{Std} = \sqrt{4.24}$$

$$= 2.06$$

Figure: Illustration of a problem solved by the student

Reflective Critique:

Conclude the activity with a class discussion, encouraging groups to share their insights and any surprising discoveries about the assumptions underlying their perspectives on numerical summaries (e.g., mean, median, standard deviation) and data interpretation.

- **Discussing with students:**

- ❖ Students expressed appreciation for the collaborative problem-solving aspect, noting that working together on real datasets helped clarify their understanding of numerical summaries.
- ❖ The calculation and interpretation process revealed diverse perspectives on how summary statistics like mean and standard deviation influence data analysis, encouraging critical dialogue about their assumptions and significance in EDA.

- **Benefit of the practice:**

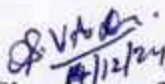
- ❖ Improved clarity on the application and interpretation of numerical summaries in real-world datasets.
- ❖ Enhanced ability to articulate positions based on calculated evidence and shared assumptions about data variability and central tendency.

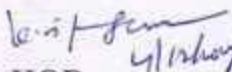
- **Challenges faced in implementation:**

- ❖ Some groups required additional guidance on how to interpret summary statistics in the context of data variability and skewness.
- ❖ Time constraints made it difficult for some groups to thoroughly discuss the implications of their results.
 - difficult for some groups to complete the sorting process thoroughly.

References:

- <https://teaching.unl.edu/resources/lecture-reaction/>
<https://www.skillshub.com/blog/active-learning-strategies-examples/>
<https://www.prodigygame.com/main-en/blog/active-learning-strategies-examples/>


Signature of Faculty Member


HOD

- Subtract the mean from each number, square the result, and find the average: $(5-15)^2+(10-15)^2+(15-15)^2+(20-15)^2+(25-15)^2=(-10)^2+(-5)^2+0^2+5^2+10^2=100+25+0+25+100=250$.
- Then, divide by the number of values (5): $250 / 5 = 50$.
- So, the variance is 50.

Step 3: Calculate the Standard Deviation

- Take the square root of the variance: $\sqrt{50} \approx 7.07$.
- So, the standard deviation is approximately 7.07.

5. Consider the dataset: 11, 13, 15, 17, 19.

Step 1: Calculate the Mean

- Add up all the numbers: $11 + 13 + 15 + 17 + 19 = 75$.
- Divide by the number of values (5): $75 / 5 = 15$.
- So, the mean is 15.

Step 2: Calculate the Variance

- Subtract the mean from each number, square the result, and find the average: $(11-15)^2+(13-15)^2+(15-15)^2+(17-15)^2+(19-15)^2=(-4)^2+(-2)^2+0^2+2^2+4^2=16+4+0+4+16=40$.
- Then, divide by the number of values (5): $40 / 5 = 8$.
- So, the variance is 8.

Step 3: Calculate the Standard Deviation

- Take the square root of the variance: $\sqrt{8} \approx 2.83$.
- So, the standard deviation is approximately 2.83.

Practice Problems

1. Find the mean, variance, and standard deviation for the dataset: 10, 15, 20, 25, 30.
2. Calculate the mean, variance, and standard deviation for: 5, 10, 15, 20, 25.
3. Determine the mean, variance, and standard deviation for: 4, 6, 8, 10, 12, 14.
4. Find the mean, variance, and standard deviation for the dataset: 1, 4, 9, 16, 25.
5. Calculate the mean, variance, and standard deviation for: 3, 6, 9, 12, 15.
6. Determine the mean, variance, and standard deviation for: 8, 16, 24, 32, 40.
7. Find the mean, variance, and standard deviation for the dataset: 2, 4, 6, 8, 10, 12.
8. Calculate the mean, variance, and standard deviation for: 11, 22, 33, 44, 55.
9. Determine the mean, variance, and standard deviation for: 7, 14, 21, 28, 35.
10. Find the mean, variance, and standard deviation for the dataset: 13, 26, 39, 52, 65.

Examples

1. Consider the data set: 4, 8, 6, 5, 3, 9.

Step 1: Calculate the Mean

- Add up all the numbers: $4 + 8 + 6 + 5 + 3 + 9 = 35$.
- Divide by the number of values (6): $35 / 6 = 5.83$.
- So, the mean is 5.83.

Step 2: Calculate the Variance

Subtract the mean from each number, square the result, and find the average:

- $(4 - 5.83)^2 + (8 - 5.83)^2 + (6 - 5.83)^2 + (5 - 5.83)^2 + (3 - 5.83)^2 + (9 - 5.83)^2$
- $= 3.35 + 4.68 + 0.03 + 0.69 + 8.00 + 10.03 = 26.78$.
- Then, divide by 6: $26.78 / 6 = 4.80$.
- So, the variance is 4.80.

Step 3: Calculate the Standard Deviation

- Take the square root of the variance: $\sqrt{4.80} = 2.19$.
- So, the standard deviation is 2.19.

2. Consider the dataset: 2, 4, 6, 8, 10.

Step 1: Calculate the Mean

- Add up all the numbers: $2 + 4 + 6 + 8 + 10 = 30$.
- Divide by the number of values (5): $30 / 5 = 6$.
- So, the mean is 6.

Step 2: Calculate the Variance

- Subtract the mean from each number, square the result, and find the average: $(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2 = (-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2 = 16 + 4 + 0 + 4 + 16 = 40$.
- Then, divide by the number of values (5): $40 / 5 = 8$.
- So, the variance is 8.

Step 3: Calculate the Standard Deviation

- Take the square root of the variance: $\sqrt{8} \approx 2.83$.
- So, the standard deviation is approximately 2.83.

3. Consider the dataset: 3, 7, 7, 19, 24.

Step 1: Calculate the Mean

- Add up all the numbers: $3 + 7 + 7 + 19 + 24 = 60$.
- Divide by the number of values (5): $60 / 5 = 12$.
- So, the mean is 12.

Step 2: Calculate the Variance

- Subtract the mean from each number, square the result, and find the average: $(3-12)^2 + (7-12)^2 + (7-12)^2 + (19-12)^2 + (24-12)^2 = (-9)^2 + (-5)^2 + (-5)^2 + 7^2 + 12^2 = 81 + 25 + 25 + 49 + 144 = 324$.
- Then, divide by the number of values (5): $324 / 5 = 64.8$.
- So, the variance is 64.8.

Step 3: Calculate the Standard Deviation

- Take the square root of the variance: $\sqrt{64.8} \approx 8.05$.
- So, the standard deviation is approximately 8.05.

4. Consider the dataset: 5, 10, 15, 20, 25.

Step 1: Calculate the Mean

- Add up all the numbers: $5 + 10 + 15 + 20 + 25 = 75$.
- Divide by the number of values (5): $75 / 5 = 15$.
- So, the mean is 15.

Step 2: Calculate the Variance